

```
!pip install opencv-python-headless scikit-image matplotlib numpy
```

```
Requirement already satisfied: opencv-python-headless in c:\users\shriy\anaconda3\lib\site-packages (4.12.0.88)
Requirement already satisfied: scikit-image in c:\users\shriy\anaconda3\lib\site-packages (0.24.0)
Requirement already satisfied: matplotlib in c:\users\shriy\anaconda3\lib\site-packages (3.10.7)
Requirement already satisfied: numpy in c:\users\shriy\anaconda3\lib\site-packages (2.2.6)
Requirement already satisfied: scipy>=1.9 in c:\users\shriy\anaconda3\lib\site-packages (from scikit-image) (1.16.3)
Requirement already satisfied: networkx>=2.8 in c:\users\shriy\anaconda3\lib\site-packages (from scikit-image) (3.5)
Requirement already satisfied: pillow>=9.1 in c:\users\shriy\anaconda3\lib\site-packages (from scikit-image) (12.0.0)
Requirement already satisfied: imageio>=2.33 in c:\users\shriy\anaconda3\lib\site-packages (from scikit-image) (2.37.0)
Requirement already satisfied: tifffile>=2022.8.12 in c:\users\shriy\anaconda3\lib\site-packages (from scikit-image) (2025.10.16)
Requirement already satisfied: packaging>=21 in c:\users\shriy\anaconda3\lib\site-packages (from scikit-image) (25.0)
Requirement already satisfied: lazy-loader>=0.4 in c:\users\shriy\anaconda3\lib\site-packages (from scikit-image) (0.4)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\shriy\anaconda3\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\shriy\anaconda3\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\shriy\anaconda3\lib\site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\shriy\anaconda3\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: pyparsing>=3 in c:\users\shriy\anaconda3\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\shriy\anaconda3\lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\shriy\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

```
import os
import warnings
import numpy as np
import matplotlib.pyplot as plt

import cv2

from skimage import io, feature, img_as_ubyte

warnings.filterwarnings("ignore", category=UserWarning)
```

```

def show_two_images(img1, title1, img2, title2, cmap1='gray',
cmap2='gray', figsize=(12,6)):
    plt.figure(figsize=figsize)
    plt.subplot(1,2,1)
    plt.title(title1)
    plt.axis('off')
    if img1.ndim == 2:
        plt.imshow(img1, cmap=cmap1)
    else:
        plt.imshow(cv2.cvtColor(img1, cv2.COLOR_BGR2RGB))
    plt.subplot(1,2,2)
    plt.title(title2)
    plt.axis('off')
    plt.imshow(img2, cmap=cmap2)
    plt.show

IMAGE_PATH = "sanika.jpg"

if not os.path.exists(IMAGE_PATH):
    raise FileNotFoundError(f"Image not found at '{IMAGE_PATH}'. Put
your image in the notebook folder or chnage the IMAGE_PATH.")

img_color = cv2.imread(IMAGE_PATH)
img_gray = cv2.imread(IMAGE_PATH, cv2.IMREAD_GRAYSCALE)

if img_color is None or img_gray is None:
    raise ValueError("Failed to load image.")

plt.figure(figsize=(6,6))
plt.title("Original (RGB)")
plt.axis('off')
plt.imshow(cv2.cvtColor(img_color, cv2.COLOR_BGR2RGB))
plt.show()

```

Original (RGB)



```
low_thresh = 100
high_thresh = 200

edges = cv2.Canny(img_gray, low_thresh, high_thresh)
show_two_images(img_gray, "Original (grayscale)", edges, f"Canny edges
(thr={low_thresh},{high_thresh})")
```


Original (grayscale)



Canny edges (thr=100,200)



```
image_for_lbp = img_as_ubyte(io.imread(IMAGE_PATH, as_gray=True))

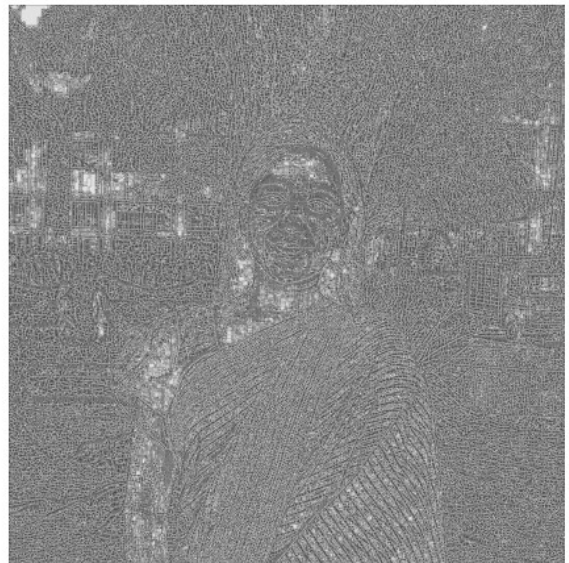
P = 8
R = 1
method = 'uniform'

lbp = feature.local_binary_pattern(image_for_lbp, P=P, R=R,
method=method)
show_two_images(image_for_lbp, "Original (for LBP)", lbp, f"LBP(P={P},
R={R}, method='{method}')
```

Original (for LBP)



LBP(P=8, R=1, method='uniform')



```
# SIFT may require opencv-contrib-python; use fallback to ORB if SIFT not present.
```

```
gray = cv2.cvtColor(img_color, cv2.COLOR_BGR2GRAY)
```

```
keypoints = None
```

```
image_with_keypoints = None
```

```
try:
```

```
    # Try SIFT
```

```
    sift = cv2.SIFT_create()
```

```
    keypoints = sift.detect(gray, None)
```

```
    image_with_keypoints = cv2.drawKeypoints(img_color, keypoints,  
None, flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS)
```

```
    print(f"Detected {len(keypoints)} SIFT keypoints.")
```

```
except Exception as e:
```

```
    print("SIFT not available or failed; using ORB instead.
```

```
(Exception: {}).format(e))
```

```
    orb = cv2.ORB_create(nfeatures=500)
```

```
    keypoints = orb.detect(gray, None)
```

```
    image_with_keypoints = cv2.drawKeypoints(img_color, keypoints,  
None, flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS)
```

```
    print(f"Detected {len(keypoints)} ORB keypoints.")
```

```
plt.figure(figsize=(8,6))
```

```
plt.title("Keypoints (SIFT or ORB)")
```

```
plt.axis('off')
```

```
plt.imshow(cv2.cvtColor(image_with_keypoints, cv2.COLOR_BGR2RGB))
```

```
plt.show()
```

```
Detected 21295 SIFT keypoints.
```

Keypoints (SIFT or ORB)

